OXFORD AQA

INTERNATIONAL QUALIFICATIONS

INTERNATIONAL AS

PSYCHOLOGY

PS02

Unit 2 Biopsychology, Development and Research Methods 1

Mark scheme

June 2025

Version: 0.1 Pre-Standardisation

256XPS02/MS

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from www.oxfordaqa.com

Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Section A: Biopsychology **Total for this section: 30 marks**

Question	Marking Guidance	Total Marks
01	**What is meant by localisation of function in the brain?** **Award 1 mark** for an outline of localisation of function. **Possible content** Localisation of function is the idea that certain functions (eg language, memory, etc) occur only in specific areas of the brain. Accept alternative wording.	**1** **AO1 = 1**

Question	Marking Guidance	Total Marks
02	**Name a localised area of the brain and outline its function.** **Award marks as follows** **1 mark** for an example of a localised area of the brain. **and** **1 mark** for an outline of its function. **Possible content** • Motor centres: to direct movement of body. • Somatosensory centres: to process and interpret information from senses. • Visual centres: to receive and process visual information. • Auditory centres: to receive and process auditory information. • Language centres: to understand and produce speech. • Broca's area: involved in language production and speech control. • Wernicke's area: involved in speech comprehension. Students may refer to other specific brain areas.	**2** **AO1 = 2**

<table>
<tr><th>Question</th><th>Marking Guidance</th><th>Total Marks</th></tr>
<tr><td>03</td><td>Using one example of a named gland, briefly explain the function of glands and hormones in the human body.

Award marks as follows
1 mark for naming a gland.

Possible content
• Adrenal medulla, pineal gland, pituitary gland, thyroid gland.

and
2 marks for a clear explanation of the function of glands and hormones.
1 mark for a muddled or vague explanation of the function of glands and hormones, or if the answer refers to either glands or hormones.

Possible content
• Glands are organs that release hormones into the blood.
• Hormones are chemicals (released from glands) that act on target structures to alter their function or to release other hormones.

Credit other relevant material.</td><td>3

AO1 = 3</td></tr>
</table>

Question	Marking Guidance	Total Marks
04	**Describe the function of the peripheral nervous system.**	4 AO1 = 4

Possible content

- It functions as an intermediary between the Central Nervous System (CNS) and the body's sensory and motor systems.
- It plays a critical role in transmitting information and regulating various physiological processes required for the body to function properly.
- It collects sensory information through sensory receptors which is then transmitted to the CNS for processing and interpretation.
- It conveys motor commands from the CNS to muscles and glands throughout the body.
- Motor commands initiate muscle contractions, leading to voluntary movements (somatic motor division) or control various involuntary processes like digestion, heart rate, and glandular secretions (autonomic motor division).

Credit other relevant material.

Level	Description	Marks
2	Knowledge of the function of the peripheral nervous system is accurate with some detail. The answer is clear with appropriate use of specialist terminology.	**3–4**
1	Knowledge of the function of the peripheral nervous system is limited, vague or muddled. Specialist terminology is either absent or inappropriately used.	**1–2**
0	No creditable content.	**0**

Question	Marking Guidance	Total Marks
05	**Explain one strength of split-brain research.** **Possible content** • Highly controlled experiment using standardised procedures, eg Sperry controlled time of exposure to stimulus, distance to screen, projection of stimulus – results, providing convincing support for hemispheric lateralisation (good internal validity). • Due to the high levels of control, the research is easily replicable. Subsequent studies have found similar results, eg Gazzaniga (2005), Turk et al. (2002), suggesting that the results from split-brain research have good reliability. • The technique has been adapted in order to research neurotypical patients. This research confirmed the right visual field advantage for words and left visual field advantage for visuospatial stimuli, suggesting that results using split-brain patients can be generalised to a neurotypical target population. • Sperry's research has enabled an advancement of knowledge. The findings were ground-breaking and provided a unique insight into the functioning of the brain while delivering scientific evidence for hemispheric lateralisation. Credit any other relevant material.	**4** **AO3 = 4**

Level	Description	Marks
2	Explanation of one strength of split-brain research is accurate with some detail. The answer is clear with appropriate use of specialist terminology.	**3–4**
1	Explanation of one strength of split-brain research is limited, vague or muddled. Specialist terminology is either absent or inappropriately used.	**1–2**
0	No creditable content.	**0**

<table>
<tr><th>Question</th><th>Marking Guidance</th><th>Total Marks</th></tr>
<tr><td>06</td><td>

Describe the procedure and findings of one study into plasticity.

Possible content

- Blakemore & Mitchell (1973) investigated the development of the visual cortex in cats. Kittens were reared in an environment with either black vertical or horizontal stripes. After seven and a half months, two of the kittens, one from each environment were anaesthetised and their neurophysiology was examined. All kittens showed behaviour blindness as they could not detect objects or contours that were aligned oppositely to their previous environment.
- Maguire (2000) investigated hippocampus changes in taxi drivers. Maguire took MRI scans of 16 right-handed male London taxi drivers who had been driving for more than 1.5 years. These were compared to scans of 50 healthy right-handed males who did not drive taxis. Maguire found increased grey matter in the (posterior) hippocampi of the brains of taxi drivers compared with controls. They also found a positive correlation between the amount of time spent as a taxi driver and volume in the right posterior hippocampus.

Credit other relevant material.

Note: A max of 3 marks can be awarded if **only** procedure or findings are at level 2.

Level	Description	Marks
2	Knowledge of procedure and/or findings of one study investigating plasticity is accurate with some detail. The answer is clear with appropriate use of specialist terminology.	3–4
1	Knowledge of procedure and/or findings of one study investigating plasticity is limited, vague or muddled. Specialist terminology is either absent or inappropriately used.	1–2
0	No creditable content.	0

</td><td>4

AO1 = 4</td></tr>
</table>

<table>
<tr><th>Question</th><th>Marking Guidance</th><th>Total Marks</th></tr>
<tr><td>07</td><td>Mariam is pouring hot water into her cup. She pours some water over her fingers and quickly drops the cup.

Using your knowledge of the functions of sensory neurons and relay neurons and motor neurons, explain why Mariam drops the cup.

Award marks as follows
2 marks for a clear application of how the sensory neurons are involved in dropping the cup.
1 mark for a muddled or vague application of how the sensory neurons are involved in dropping the cup.

Possible content
The receptors in Mariam’s fingers would sense the heat from the hot water and the sensory neurons send that information via the PNS to the spinal cord/CNS (reflex arc).

And
2 marks for a clear application of how the relay neurons are involved in dropping the cup.
1 mark for a muddled or vague application of how the relay neurons are involved in dropping the cup.

Possible content
Relay neurons would be involved in analysing the sensation of heat and pain caused by the hot water. They initiate the motor response, removing Mariam’s fingers from the heat source, eg the hot water.

And
2 marks for a clear application of how the motor neurons are involved in dropping the cup.
1 mark for a muddled or vague application of how the motor neurons are involved in dropping the cup.

Possible content
The message from the spinal cord instructs the motor neurons in Mariam’s fingers to get away from the heat source, she relaxes the fingers and drops the cup.

Credit other relevant material.</td><td>6

AO2 = 6</td></tr>
</table>

<table>
<tr><th>Question</th><th>Marking Guidance</th><th>Total Marks</th></tr>
<tr><td>08</td><td>

Enrico is 4 years old and has just had brain surgery. Following the surgery, Enrico is unable to move his left leg.

In a conversation with Enrico's parents, the doctor tells them that Enrico may recover some movement in his left leg.

Explain how plasticity and/or functional recovery after trauma might enable Enrico to use his leg again.

Possible content

- Functional recovery is the recovery of abilities that have been lost after brain trauma like surgery.
- Enrico's brain is likely to compensate for loss of function and he should regain some functioning of his left leg.
- As Enrico is quite young (4 years old) his brain is more plastic.
- Plasticity enables Enrico's brain to recover function due to: extra stimulation of the surrounding undamaged areas; axon sprouting; any remaining axons becoming super-sensitive.
- The extent of Enrico's recovery is dependent on Enrico's rehabilitation programme, his efforts and his perseverance in performing certain exercises. Stimulating the affected brain area will help the recovery of functioning.

Credit other relevant material.

<table>
<tr><th>Level</th><th>Description</th><th>Marks</th></tr>
<tr><td>3</td><td>Application of plasticity and/or functional recovery to Enrico is detailed and appropriate. The answer is clear with appropriate use of specialist terminology.</td><td>5–6</td></tr>
<tr><td>2</td><td>Application of plasticity and/or functional recovery to Enrico is relevant but detail is lacking. The answer lacks clarity in places. Specialist terminology is occasionally used appropriately.</td><td>3–4</td></tr>
<tr><td>1</td><td>Application of plasticity and/or functional recovery to Enrico is very limited. The answer is vague/muddled. Specialist terminology is either absent or inappropriately used.</td><td>1–2</td></tr>
<tr><td>0</td><td>No creditable content.</td><td>0</td></tr>
</table>

</td><td>6

AO2 = 6</td></tr>
</table>

Section B: Cognitive Development **Total for this section: 30 marks**

Question	Marking Guidance	Total Marks
09	**Which of the following terms is not part of Piaget's theory of cognitive development?** **Shade one box only.** **Answer key:** **D** – Scaffolding	1 AO1 = 1

Question	Marking Guidance	Total Marks
10	**Describe how Piaget explains schema development through the processes of assimilation and accommodation.** **Award marks as follows** **3 marks:** The description of how schemas develop through the processes of assimilation and accommodation is detailed. The answer is clear with appropriate use of specialist terminology. **2 marks:** The description of how schemas develop through the processes of assimilation and/or accommodation lacks detail. The answer lacks clarity in places. **1 mark:** The description of how schemas develop through the processes of assimilation and/or accommodation is briefly presented. The answer is very limited/vague/muddled. **Possible content** • Individuals initially use an existing schema to make sense of the world. Through the process of assimilation, they incorporate new information or experiences into the existing schema. • When faced with new, incompatible information, they must adjust their schema through accommodation. This means the schema needs to be changed or updated in the light of this new experience. • This ongoing process of assimilation and accommodation leads to the development of a more complex and sophisticated cognitive structure (schema). Credit examples of assimilation and accommodation as part of the description. Credit other relevant material.	3 AO1 = 3

Question	Marking Guidance	Total Marks
11	**Evaluate research investigating the role of the mirror neuron system in social cognition.** **Possible evaluation** • Using animals allows the use of invasive techniques such as implanting electrodes into the brain of monkeys: Di Pellegrin et al. (1992), Rizzolatti et al. (1996), in order to measure activity of individual neurons which would be unethical to do with humans. • Animals such as monkeys have mirror systems but their social cognition is less developed compared to humans whose systems are more evolved, eg an area which is less developed in monkeys is deception. This questions the ability to generalise results from animal research. • Use of fMRI scans, eg fMRI scanner of finger movement, Iacoboni (1999) provide objective and scientific evidence, demonstrating that discoveries from animal research are also relevant for humans. • fMRI scans are correlational evidence. It remains unclear if the mirror neurons were activated by simply watching someone else move a finger. • Research has confirmed the existence of this type of neuron system in the brain. Credit other relevant material.	**6** **AO3 = 6**

Level	Description	Marks
3	Evaluation of research investigating mirror neurons is effective. The answer is clear with appropriate use of specialist terminology.	5–6
2	Evaluation of research investigating mirror neurons is relevant but effectiveness is lacking. The answer lacks clarity in places. Specialist terminology is occasionally used appropriately.	3–4
1	Evaluation of research investigating mirror neurons is very limited. The answer is vague and/or muddled. Specialist terminology is either absent or inappropriately used.	1–2
0	No creditable content.	0

<table>
<tr><th>Question</th><th>Marking Guidance</th><th>Total Marks</th></tr>
<tr><td>12</td><td>
Yonda is 6 months old and takes part in a violation of expectation study. She is shown a truck moving along a track and disappearing into a tunnel.

There are two conditions in the study.

Condition 1: the truck enters the tunnel and can be seen coming out of the tunnel.

Condition 2: the truck enters the tunnel but does not come out.

The researcher found that Yonda looked at the tunnel for longer in Condition 2 than in Condition 1.

Describe and evaluate Baillargeon’s violation of expectation research. In your answer, refer to Yonda’s responses in the study.

Possible content
• An experimental method of assessing infants’ understanding of the physical world.
• Use of eye tracking technology to measure how long the child looks at the object/event.
• Based on visual preference technique assuming that the longer an infant looks at a stimulus, the more interested/less familiar it is to the child.
• 2 stages of violation of expectations (VOE):
1. Habituation to the expected event – child is repeatedly exposed to a particular event so that it stops responding to it, eg a train goes into a tunnel and re-emerges.
2. For the second part children are divided up into two groups: group 1 watches the expected event and group 2 watches an unexpected event (eg train goes into tunnel and does not re-emerge or another object re-emerges instead) which violates the child’s expectation.
• Description of research by Baillargeon et al. (1985) using the drawbridge and hidden blocks of wood; resting object research.
• Description of different events investigated: occlusion, containment, covering and support.
• Theory of innate object knowledge and innate fast learning.

Possible application
• In Condition 1, Yonda is shown the expected (possible) event. The truck enters the tunnel and appears again. Yonda expected this to happen and for this reason, Yonda did not spend much time looking at it as her expectations were not violated.
• Condition 2 is the unexpected (impossible) event because the truck should have appeared again but it did not. This violated her expectations. Yonda was surprised in the second condition and this caused Yonda to look longer at the tunnel.
</td><td>20

AO1 = 8
AO2 = 4
AO3 = 8</td></tr>
</table>

Possible evaluation
- Improved method researching understanding of children compared to Piaget – showing that children as young as 3.5 months old understand object permanence without needing to be able to reach for objects.
- Eye tracking technology measures – objective and scientific measurement.
- Use of inference assuming that children look at something because it violates their expectation.
- Alternative explanation: preference for full circle (Bogartz/Rivera) or presence of an afterimage of blocks (Haith).
- Research support eg by Kaufman et al. using brain scans.
- Influential work that has led to the development of several connectionist models of brain development.

Credit other relevant material.

Level	Description	Marks
4	Knowledge of Baillargeon's violation of expectation research is accurate and generally well detailed. Evaluation is effective. Application is appropriate. Minor detail and/or expansion of argument is sometimes lacking. The answer is clear and focused. Specialist terminology is used effectively.	**15–20**
3	Knowledge of Baillargeon's violation of expectation research is evident but there are occasional inaccuracies and/or omissions. There is some effective evaluation. There is some appropriate application. The answer is mostly clear and organised but occasionally lacks focus. Specialist terminology is used appropriately.	**11–14**
2	Limited knowledge of Baillargeon's violation of expectation research is present. Any evaluation/application is of limited effectiveness. The answer lacks clarity, accuracy and organisation in places. Specialist terminology is occasionally used appropriately.	**6–10**
1	Knowledge of Baillargeon's violation of expectation research is very limited. Evaluation/application is limited, poorly focused or absent. The answer as a whole lacks clarity, has many inaccuracies and is poorly organised. Specialist terminology is either absent or inappropriately used.	**1–5**
0	No creditable content.	**0**

Section C: Research Methods 1 **Total for this section: 30 marks**

	A psychologist aims to investigate the possible relationship between playing sports and mathematical ability. For her study, she uses 20 tennis players from the local tennis club. **To collect the data for her co-variables, the psychologist asks every tennis player how many years they have been playing tennis.** **She also gives each of the 20 tennis players a mathematics test with a maximum score out of 50.**	

Question	Marking Guidance	Total Marks
13	**What is meant by co-variables?** **Award 1 mark:** co-variables are the behaviours/qualities/abilities measured or used in a correlational analysis. Accept alternative wording.	**1** **AO1 = 1**

Question	Marking Guidance	Total Marks
14	**Identify the co-variables in this study.** **Award 1 mark** for each correctly identified co-variable: • the number of years of playing tennis • the score (out of 50) in the mathematics test.	**2** **AO2 = 2**

Question	Marking Guidance	Total Marks
15	**What is meant by the hypothesis in a study?** **Award marks as follows** **2 marks** for a clear and effective definition of a hypothesis. **1 mark** for a limited/vague/muddled definition of a hypothesis. **Possible content** • A testable statement about the outcome of an investigation, using clearly operationalised variables. Accept alternative wording.	**2** **AO1 = 2**

Question	Marking Guidance	Total Marks
16	**Write an appropriate non-directional hypothesis for this study.** **Award marks as follows** **3 marks** for an appropriate non-directional hypothesis, with both co-variables fully operationalised. **2 marks** for a non-directional hypothesis with both co-variables, but lacking clarity or where only one co-variable is operationalised. **1 mark** for a muddled non-directional hypothesis with both co-variables but neither operationalised. **Note:** no credit for a hypothesis stating a difference or a directional hypothesis or a null hypothesis. **Possible content** There will be a relationship/correlation between the number of years a person plays tennis and their score (out of 50) in the mathematics test. Accept alternative wording.	**3** **AO2 = 3**

Question	Marking Guidance	Total Marks
17	**Explain why it is important to consider ethics in the design and conduct of psychological studies.** **Award marks as follows** **3 marks:** The explanation of why it is important to consider ethics is detailed. The answer is clear with appropriate use of specialist terminology. **2 marks:** The explanation of why it is important to consider ethics lacks detail. The answer lacks clarity in places. **1 mark:** The explanation of why it is important to consider ethics is briefly presented. The answer is very limited/vague/muddled. **Possible content** • Ethics are important in psychological research to avoid directly or indirectly harming participants. • They ensure that participants are treated with respect. The participants should leave any study feeling no worse about themselves than before their participation. • They promote the integrity of research and they uphold the credibility and trustworthiness of the scientific community. • If participants feel harmed, they might take the psychologist to court or refuse in future to take part in any research. Credit other relevant material.	**3** **AO1 = 3**

<table>
<tr><th>Question</th><th>Marking Guidance</th><th>Total Marks</th></tr>
<tr><td>18</td><td>
Consent is an ethical consideration.
Explain what information the psychologist should provide to the tennis players to enable them to give their consent to take part in the study.
Possible content

The psychologist should tell them about the aim of the study which is to investigate the relationship between playing tennis and their score in mathematics.
The psychologist should inform the tennis players that the participation is voluntary and that they have the right to withdraw themselves at any point and withdraw their data at the end of the study.
She should guarantee the confidentiality and anonymity of the tennis player’s data, and the privacy of their personal information.

Credit other relevant material.
<table>
<tr><th>Level</th><th>Description</th><th>Marks</th></tr>
<tr><td>2</td><td>Knowledge of consent applied to the correlational study is accurate with some detail. The answer is clear with appropriate use of specialist terminology.</td><td>3–4</td></tr>
<tr><td>1</td><td>Knowledge of consent applied to the correlational study is limited, vague or muddled. Specialist terminology is either absent or inappropriately used.</td><td>1–2</td></tr>
<tr><td>0</td><td>No creditable content.</td><td>0</td></tr>
</table>
</td><td>4

AO2 = 4</td></tr>
</table>

<table>
<tr><th>Question</th><th>Marking Guidance</th><th>Total Marks</th></tr>
<tr><td>19</td><td>
What is meant by a positive correlation?
Award marks as follows
2 marks for a clear definition of a positive correlation.
1 mark for a muddled or vague definition of a positive correlation.
Possible content
A positive correlation means when one variable increases in value, the other variable also increases in value.
Credit other wording.
</td><td>2

AO1 = 2</td></tr>
</table>

	The psychologist decides to do a follow up experiment with the same 20 tennis players to see if playing tennis has a direct effect on the ability to solve mathematical questions. **For this experiment, she paired up the tennis players on their mathematical ability by using their scores on the mathematics test they had done previously.** **Each tennis player takes part in one of the following conditions.** **Condition 1: playing tennis for 60 minutes.** **Condition 2: sitting quietly for 60 minutes.** **After the 60 minutes, the tennis players were asked to answer a new mathematical test.**	

Question	Marking Guidance	Total Marks
20	**Identify the experimental design used in this study. Briefly explain your answer.** **Award marks as follows** **1 mark** for matched pairs design. **And 1 further mark** for a brief explanation: tennis players were matched/paired using their mathematical score from their previous mathematics test. Credit alternative wording.	**2** **AO2 = 2**

Question	Marking Guidance	Total Marks
21	**Describe how the psychologist could have randomly allocated the players to the two conditions of the experiment.** **Award 1 mark for each point** • The psychologist could prepare slips of paper for each pair with a number 1 and a number 2 on it and ask the two tennis players to choose/pick one slip of paper. • The tennis player who draws number 1 is allocated to Condition 1 and the player who draws number 2 is allocated to Condition 2. • This process is repeated for all matched pairs until all players have been allocated to one condition. Credit alternative wording. Credit alternative approaches, such as the use of a computer to randomly allocate the names to each condition.	**3** **AO2 = 3**

Question	Marking Guidance	Total Marks
22	**Identify the fully operationalised independent variable of this experiment.** **Award marks as follows** **2 marks** for identifying the fully operationalised independent variable. **1 mark** for identifying a muddled or vague explanation of the independent variable. **Content** The independent variable is whether they played tennis (for 60 minutes) or sat quietly (for 60 minutes).	**2** **AO2 = 2**

Question	Marking Guidance	Total Marks
23	**Identify one extraneous variable in the follow up experiment. Briefly explain how this extraneous variable might affect the results.** **Award marks as follows** **1 mark** for a suitable extraneous variable. **And 1 further mark** for a brief explanation of how this extraneous variable might affect the results. **Possible content** • Age: younger players might still be studying mathematics and therefore be better in the mathematics test. • Occupation: players with jobs involving high levels of mathematics will do better in the mathematics test. • Gender: some research suggests there is a difference in mathematics ability between the genders, and one gender might perform better in the mathematics test. • Level of mathematics education: some may have studied mathematics for longer or at an advanced level and might therefore perform better in the mathematics test. Credit other relevant material.	**2** **AO2 = 2**

Question	Marking Guidance	Total Marks
24	**What is meant by demand characteristics?** **Award marks as follows** **2 marks** for a clear definition of demand characteristics. **1 mark** for a muddled or vague definition of demand characteristics. **Possible content** • Demand characteristics are the cues the participant uses to work out the aims of the research. • They can affect what the participants think and how they behave. • They will become confounding variables if they have affected or changed the behaviour of a participant as the results will be due to these behavioural changes. Credit an example, eg Hawthorne effect. Accept alternative wording.	**2** **AO1 = 2**

Question	Marking Guidance	Total Marks
25	**Briefly explain how demand characteristics could have affected this study.** **Award marks as follows** **2 marks** for a clear explanation of how demand characteristics could have affected the study. **1 mark** for a muddled or vague explanation of how demand characteristics could have affected the study. **Possible content** • The psychologist used tennis players from her local tennis club, therefore she might know them which might affect how they perform on the mathematics test. • The players might have not been truthful or they had forgotten when they were asked about the number of years they have been playing tennis. Credit other relevant material.	**2** **AO2 = 2**

PS02 grid

	AO1	AO2	AO3	Total
Section A				
01	1			**1**
02	2			**2**
03	3			**3**
04	4			**4**
05			4	**4**
06	4			**4**
07		6		**6**
08		6		**6**
Section B				
09	1			**1**
10	3			**3**
11			6	**6**
12	8	4	8	**20**
Section C				
13	1			**1**
14		2		**2**
15	2			**2**
16		3		**3**
17	3			**3**
18		4		**4**
19	2			**2**
20		2		**2**
21		3		**3**
22		2		**2**
23		2		**2**
24	2			**2**
25		2		**2**
Unit total	**36**	**36**	**18**	**90**